State-Dependent Macrophage Transitions in Osteoarthritis (OA) Synovitis: A Regulatory Network Model (RNM)

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Introduction

Synovial macrophages (MC) regulate joint homeostasis but can become dominant drivers of synovitis and tissue degeneration in OA [1]. Their behaviour is dynamic and multistable, shifting between inflammatory (M1-like) and resolving (M2-like) states in response to local cues. Despite expanding knowledge in synovitis and OA, linking regulatory complexity to MC state transitions and measurable biomarkers remains challenging. Here, we develop a companion diagnostic (CDx) tool based on an RNM to predict MC polarization dynamics and stratify synovitis-driven OA.

Materials and Methods

An RNM with >120 molecular nodes and >300 activating/inhibitory interactions were reconstructed from >100 experimental studies, integrating immune signaling and metabolic reprogramming. Interactions were curated primarily from human and animal macrophage and monocyte studies, with a few drawn from other cell types. Network dynamics were simulated using the Standardized Qualitative Dynamical Systems formalism [2], in which normalized node activities evolve according to , representing integrated regulatory inputs and decay. We ran 10^4 simulations with random initial node activations (0–1) to identify attractors. The resulting attractors were then perturbed by clamping selected molecules to 1 (activation) or 0 (suppression) to examine inflammation and resolution. Model outputs were compared with synovial-fluid concentrations measured by Luminex in 36 women with primary knee OA (KL grades 2–3), after log transformation and normalization of the measured values.

Results and Discussion

Basal simulations yielded M2-like (71.7%), M1-like (26.7%), and unsettled (1.6%) outcomes (Fig. 1A). Unsettled outcomes met neither classification criterion and tended toward M1-like on further simulation. Perturbation effects depended on the clamped node and starting state (Fig. 1B). From random starts (Fig. 1B1), TNF activation and IL-10 or TGF-β suppression increased M1-like, whereas activation of IL-1β, IFN-γ, NF-κB, and others favoured unsettled. The established M2-like attractor (Fig. 1B2) was more resistant: only TNF activation shifted it to M1-like. Rescue was also state dependent. AMPK, TGF-β, or CREB activation and AP-1 suppression restored M2-like more readily from unsettled (Fig. 1B4) than from established M1-like (Fig. 1B3); IL-10 activation and TNF suppression restored M2-like from both.

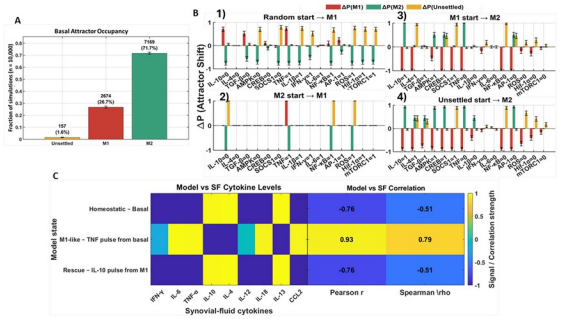


Figure 1. Basal attractors (A), state-dependent perturbation responses (B), and model–synovial fluid cytokine concordance (C).

Network architecture offers a mechanistic explanation for these differential effects. TNF has less constrained access to NF-κB/AP-1 through TRAF2 and suppresses the resolving axis, enabling it to overcome M2 stability. In contrast, IL-1β signalling through IRAK–TRAF6 is gated by negative feedback and buffered by the IL-1β–PGE_2–CREB–IL-10/IL1RA resolving branch. IFN-γ induces SOCS1, limiting its signalling at JAK. IL-10 inhibits inflammatory regulators while reinforcing resolution; IL-4 promotes the M2 transcriptional program with limited inflammatory repression. TGF-β signals through suppressive SMAD2/3 and inflammatory TAK1 branches, consistent with context-dependent effects. Consistent with TNF's predicted inflammatory influence, its induced model cytokine profile showed the strongest agreement with the OA synovial-fluid profile (Pearson r = 0.93; Spearman ρ = 0.79; Fig. 1C), supporting the relevance of the predicted inflammatory state. The RNM links regulatory mechanisms to MC behaviour, enabling systematic exploration of immune-related therapeutic targets in OA.

References

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